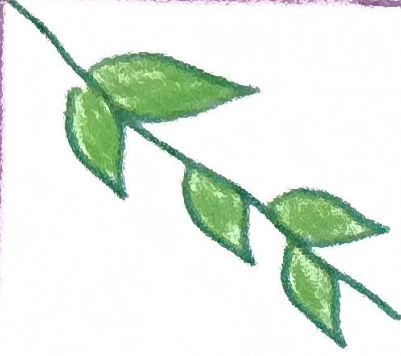




Name: M. Nivetha

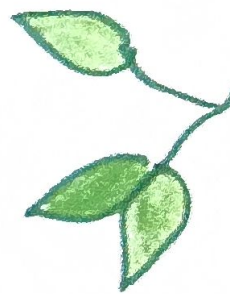
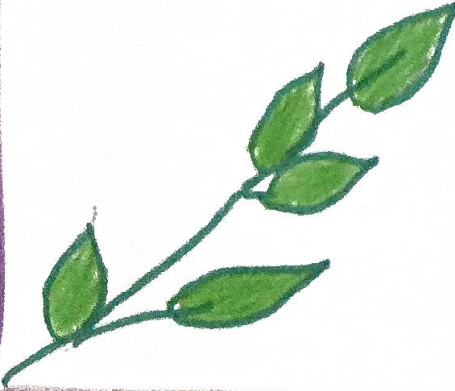
RegNo: 192471040

Course Code: ITA0611

Course: Machine Learning

Date: 8/10/2026

Assessment: Col-AT2-Concept Mapping



Machine Learning Concept Map

1. What is Machine Learning?

- A branch of Artificial Intelligence (AI)
- Enables Computers to learn from data without being explicitly programmed.
- Learns patterns and make predictions or decisions

2. Type of Machine Learning

Supervised Learning

- Labeled data
- Classification
- Regression (Ex: Spam detection)

Unsupervised Learning

- Unlabeled data
- Clustering
- Association (Ex: Customer Segmentation)

Reinforcement Learning

- Learns by rewards
- Trial and error
- Decision making (Ex: Robotics, Game)

3. Learning Techniques / Algorithms

- Decision Tree
- Naive Bayes
- K-Nearest Neighbour (K-NN)
- Support Vector Machine (SVM)
- Linear Regression
- Neural Network / Deep Learning

4. Learning process

Data Collection

→ Gather relevant data from various source.

Data preprocessing

→ Clean, handle missing value, normalize

Feature Extraction

→ Select important features

Model Training

→ learn patterns from data

Evaluation

→ Test performance using metrics

6. Evaluation Metrics

- Accuracy - overall correct predictions
- Precision - correct positive prediction
- Recall - find all positive cases
- F1 Score - Balance of precision and recall

	Predicted	
	Positive	Negative
Positive	TP	FN
Negative	FP	TN

Machine Learning

7. Application

- Healthcare (disease prediction)
- Finance (fraud detection)
- Education (personalized learning)
- E-Commerce (recommendation)
- Autonomous Vehicles

5. Key Concepts

Hypothesis

→ possible learned solution

Training Set

→ Pairs to model

Test Set

→ tests the model

Overfitting

→ learns noise

Underfitting

→ Too Simple

Bias & Variance

→ Model Complexity, overfitting

8. Importance & Challenges

Importance

- Better decision
- Save time
- Automate tasks

Challenges

- High Computation
- Data bias

Machine Learning System for Disaster Management

1. Data Collection

Historical Data

past flood/earthquake records
Rainfall, river levels
Soil moisture
Seismic data
Weather patterns

Real-time Data

Live sensor data
Satellite & radar data
GPS/IoT devices
Social media/news feeds

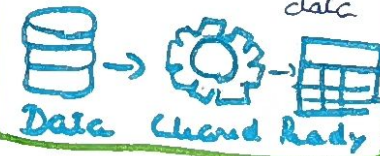
2. Model Choice

Hybrid Model (Time Series + Deep Learning)

LSTM
(Long short term Memory)
for sequential data (Ex: -
Satellite, rainfall maps)
Handles complex patterns
and improves prediction accuracy

3. Data Preprocessing

Handle missing
Value (Interpolation
mean/median)
Remove Noise and
Outliers
Normalize/Standardize data



4. Feature Selection

Identify the most
relevant features
for prediction
Using Correlation,
Chi-Square and
Domain Knowledge

Selected Rainfall,
Feature: river,
level, Soil moisture

a) Steps for Training, Validation and Evaluation of model

1. Training

Split data
(train/val/test)
Feed preprocessed
data
Optimize model
parameters
(Ex: Adam, learning rate)

2. Validation

Use validation
set
Tune hyperparam-
eters
Prevent overfitting
Check for
generalization

3. Evaluation

Test on unseen
data
Metrics: Accuracy,
Precision, Recall,
F1-Score
For regression:
MAE, RMSE

b) Early Warning and Emergency Response

Prediction (Flood/Earthquake)

Early Warning System

Alerts (SMS, app, siren)
Risk level & affected
areas
Real-time dash-
boards

Emergency Response

→ Evacuate
Vulnerable
Areas
→ Send
resources

Outcomes & Benefits

Reduces damage to property and Infrastructure
Enables faster and better decision-making

Concept Map - Find S Algorithm

1. Learning Problem

- Supervised learning (classification)
- Find a maximally specific hypothesis from training data.

2. Representation

- Hypothesis: attribute-value pair
- Most specific Hypothesis h

3. Algorithm Overview

- Initialize h to most specific value
- For each positive value
 - Compare with h
 - Generalize h where needed

4. Dataset

Ex	Outlook	Temperature	Humidity	Wind	play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Overcast	Mild	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No

= Positive examples (used for updates)
 = Negative examples (Ignore)

Find-S Algorithm
 Finds the most specific hypothesis consistent with positive examples

5. Flow of the Algorithm

Initialize $h = (\phi, \phi, \phi, \phi)$
 (i.e. most specific value)
 ↓
 For each positive example
 ↓
 Generalize h where needed
 ↓
 Return final h

6. Step by step updates (positive examples)

Step 1 (Ex. 3)
 (Overcast, Hot, High, Weak)
 $h_1 = \langle \text{Overcast, Hot, High, Weak} \rangle$
Update:
 From $\langle \phi, \phi, \phi, \phi \rangle$ to $\langle \text{Overcast, Hot, High, Weak} \rangle$

Step 2 (Ex. 4)
 (Rainy, Mild, High, Weak)
 $h_2 = \langle ?, \text{Mild, High, Weak} \rangle$
Update:
 Generalize Outlook

Step 3 (Ex. 5)
 (Rainy, Cool, Normal, Weak)
 $h_3 = \langle ?, \text{Cool, Normal, Weak} \rangle$
Update:
 Generalize temp from Mild to ?

Step 4 (Ex. 7)
 (Overcast, Mild, Normal, Strong)
 $h_4 = \langle ?, \text{Mild, Normal, ?} \rangle$
Update:
 Generalize Wind from Weak to ?

Final Hypothesis

$h = \langle ?, \text{Mild, Normal, ?} \rangle$
 ↳

- Outlook: Any Value
- Temp: Mild
- Humidity: Normal

To Find-S Algorithm the most specific Hypothesis consistent with all positive examples

Training Examples used for updates

Step	Ex No	Outlook	Temp	Humidity	Wind	play Tennis
1	3	Overcast	Hot	High	Weak	Yes
2	4	Rainy	Mild	High	Weak	Yes
3	5	Rainy	Cool	Normal	Weak	Yes
4	7	Overcast	Mild	Normal	Strong	Yes

Candidate Elimination Algorithm

1. Dataset

Example	Weather	Temp	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

2. Attribute Domains

Weather: {Sunny, Rainy}

Temperature: {Warm, Cold}

Humidity: {Normal, High}

Wind: {Weak, Strong}

3. Hypothesis Representation

A hypothesis is a 4-tuple:

$\langle \text{Weather}, \text{Temperature}, \text{Humidity}, \text{Wind} \rangle$

• **Specific boundary (S):** Most specific hypothesis consistent with observed positive example

• **General boundary (G):** Most general hypothesis consistent with observed negative example

Initial values: $S_0 = \langle \phi, \phi, \phi, \phi \rangle$

$G_0 = \langle ?, ?, ?, ? \rangle$

4. Step-by-step updates

Step 1: Example 1 (Yes)

Update: Initialization with the first positive example. Generalize fully, general

Result: $S_1 = \langle \text{Sunny}, \text{Warm}, \text{Normal}, \text{Weak} \rangle$, $G_1 = \phi$

Step 2: Example 2 (Yes)

Update: Generalize to all attributes that differ from Example 1.

Result: $S_2 = \langle \text{Sunny}, \text{Warm}, ?, ? \rangle$, $G_2 = \langle ?, ?, ?, ? \rangle$

Example 3 (No)

Update: Specialize into exclude the Negative example

Result: $S_3 = \langle \text{Sunny}, \text{Warm}, ?, ? \rangle$, $G_3 = \langle \text{Sunny}, ?, ?, ? \rangle$

Example 4 (Yes)

Update: Generalize to cover the new positive example

Result: $S_4 = \langle \text{Sunny}, ?, ?, \text{Weak} \rangle$, $G_4 = \langle \text{Sunny}, ?, ?, ? \rangle$

Example 5 (No)

Update: Remove the hypothesis in G that cover the Negative

Result: $S_5 = \langle \text{Sunny}, ?, ?, \text{Weak} \rangle$, $G_5 = \langle \text{Sunny}, ?, ?, ? \rangle$

5. Final Version Space

Specific Boundary (S)

$S = \langle \text{Sunny}, ?, ?, \text{Weak} \rangle$
(Most specific hypothesis)

Version Space

General Boundary (G)

$G = \langle \text{Sunny}, ?, ?, ? \rangle$
 $\langle ?, \text{Warm}, ?, ? \rangle$
Most general hypothesis

6. Interpretation

The version space contains all hypotheses consistent with the training example

S shows the most specific boundary

G shows the most general boundary

Key Takeaways

S: Most specific Consistent Hypothesis

G: Most general Consistent Hypothesis

The version space contains Hypotheses

Find - S Algorithm

1. Dataset

Ex	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

 - positive Example - negative Example

2. Find S-Algorithm

- 1) Initialize the most specific hypothesis S_0 with all attributes set to the most specific value (or empty).
- 2) For each positive example, compare with S and generalize S to include the example.
- 3) Ignore Negative Examples.
- 4) Final S is the maximally specific hypothesis.

3. Attribute Domains

Attribute	Possible Values
Age	{Young, Middle, Senior}
Income	{High, Medium, Low}
Student	{Yes, No}
Credit Rating	{Fair, Excellent}

Initial Hypothesis (most specific)
 $S_0 = \langle \phi, \phi, \phi, \phi \rangle$

4. Step-by-step updates (using only positive Example)

Step 1: Ex 1 (Ignored)

Example 1: $\langle \text{Young, High, No, Fair} \rangle \rightarrow \text{No}$

Not a positive example

No change to S .

$S_0 = \langle \phi, \phi, \phi, \phi \rangle$

Step 2: Ex 2 (Ignored)

Example 2: $\langle \text{Young, High, No, Excellent} \rangle \rightarrow \text{No}$

Not a positive example

No change to S .

$S_1 = \langle \phi, \phi, \phi, \phi \rangle$

Step 3: Ex 3 (Positive)

Example 3: $\langle \text{Middle, High, No, Fair} \rangle \rightarrow \text{Yes}$

Generalize S to include this example

$S_2 = \langle \text{Middle, High, No, Fair} \rangle$

Step 4: Ex 4 (Positive)

Example 4: $\langle \text{Senior, Medium, No, Fair} \rangle \rightarrow \text{Yes}$

Generalize S to cover this example

$S_3 = \langle ?, \text{Medium, No, Fair} \rangle$

5. Final Hypothesis

$S_{\text{final}} = \langle ?, \text{Medium, No, Fair} \rangle$

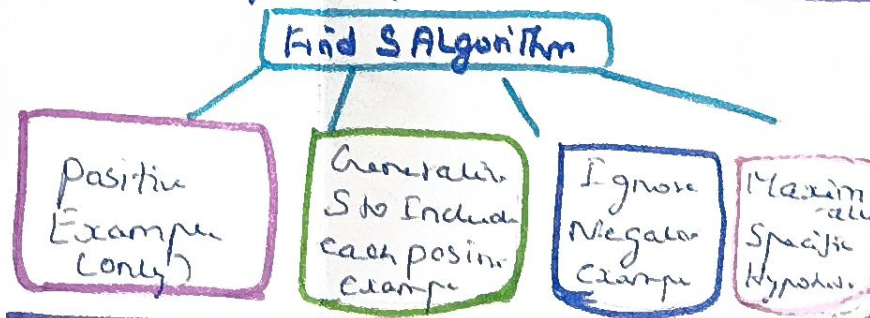
Meaning:

Age can be any value (Young, Middle, Senior)

Student = No

Income = Medium

Concept Map



Key Takeaways

Find-S uses only positive examples. It maintains the most specific hypothesis that covers all positive examples seen so far. Negative examples are ignored.